

## 45.5 Models for location, scale and shape

Quantile and expectile regression estimate one feature of the conditional law at a time and never write the law down. The opposite strategy writes it down in full: choose a parametric family with enough parameters to bend as the data require, and let every one of them depend on the covariates through its own structured additive predictor. Rigby and Stasinopoulos (2005) named the result generalized additive models for location, scale and shape.

The gain is a likelihood, and with it penalized estimation, effective degrees of freedom, AIC and Bayesian posteriors. The cost is a parametric assumption about the shape of the conditional distribution, which the data must be allowed to contradict — the subject of [Section 45.6](#).

### 45.5.1 The model

#### Definition 45.5.1. Generalized additive model for location, scale and shape

Let  $D(\theta_1, \dots, \theta_K)$  be a family of distributions on the response space with density or mass function  $d(y; \theta_1, \dots, \theta_K)$ ; in this chapter  $\theta_k$  denotes the  $k$ th parameter of that family, not the natural parameter of Definition 34.2.1. A **GAMLSS** asserts that the responses  $Y_1, \dots, Y_n$  are independent with

$$Y_i \sim D(\theta_{i1}, \dots, \theta_{iK}), \quad g_k(\theta_{ik}) = \eta_{ik}, \quad k = 1, \dots, K,$$

where each  $g_k$  is a strictly increasing **link** mapping the admissible range of  $\theta_k$  onto  $\mathbb{R}$  (the identity for an unrestricted parameter, the logarithm for a positive one, the logit for one in  $(0, 1)$ ), and each  $\eta_{ik}$  is a structured additive predictor (Definition 44.5.1)

$$\eta_{ik} = \beta_{0k} + \sum_{j=1}^{J_k} f_{jk}(v_{ij}), \quad f_{jk}(\cdot) \text{ represented by } \mathbf{Z}_{jk} \boldsymbol{\gamma}_{jk}, \quad (45.5.1)$$

with each  $\boldsymbol{\gamma}_{jk}$  carrying a quadratic penalty  $\lambda_{jk} \boldsymbol{\gamma}_{jk}^\top \mathbf{P}_{jk} \boldsymbol{\gamma}_{jk}$ . The predictors for different parameters may use different covariates and different amounts of structure.

Taking  $K = 1$  with  $D$  an exponential dispersion family and  $\eta$  linear recovers Definition 34.2.1; with additive  $\eta$  it recovers Theorem 44.6.1. The new content is  $K > 1$ : the scale, and the shape, become regression problems.

#### Example. The heteroscedastic normal model

The smallest interesting case has  $K = 2$ ,  $D = N(\mu, \sigma^2)$ , identity link on  $\mu$  and log link on  $\sigma$ :

$$Y_i \sim N(\mu_i, \sigma_i^2), \quad \mu_i = \eta_{i1}, \quad \log \sigma_i = \eta_{i2}. \quad (45.5.2)$$

The log link keeps  $\sigma_i > 0$  whatever the predictor does, as it kept a Poisson mean positive in [Chapter 37](#). With both predictors linear this is Proposition 45.1.2(b) with normal  $\varepsilon$ ; with penalized splines it is the fit of [Example \(Mean and scale together\)](#).

## 45.5.2 Estimation

### Theorem 45.5.1. Penalized likelihood and block-wise Fisher scoring

Write  $\ell_i(\boldsymbol{\theta}_i) = \log d(y_i; \theta_{i1}, \dots, \theta_{iK})$  and collect the coefficients of predictor  $k$  into  $\boldsymbol{\gamma}_k$  with design  $\mathbf{Z}_k$ , penalty matrix  $\mathbf{P}_k$  and smoothing parameter  $\lambda_k$ , so that  $\boldsymbol{\eta}_k = \mathbf{Z}_k \boldsymbol{\gamma}_k$ . The **penalized log-likelihood** is

$$\ell_p(\boldsymbol{\gamma}_1, \dots, \boldsymbol{\gamma}_K) = \sum_{i=1}^n \ell_i(\boldsymbol{\theta}_i) - \frac{1}{2} \sum_{k=1}^K \lambda_k \boldsymbol{\gamma}_k^\top \mathbf{P}_k \boldsymbol{\gamma}_k. \quad (45.5.3)$$

- (a) **Score.** With  $u_{ik} = \partial \ell_i / \partial \eta_{ik}$  and  $\mathbf{u}_k = (u_{1k}, \dots, u_{nk})^\top$ ,

$$\frac{\partial \ell_p}{\partial \boldsymbol{\gamma}_k} = \mathbf{Z}_k^\top \mathbf{u}_k - \lambda_k \mathbf{P}_k \boldsymbol{\gamma}_k. \quad (45.5.4)$$

- (b) **Fisher scoring is penalized weighted least squares.** Let  $w_{ik} = -E(\partial^2 \ell_i / \partial \eta_{ik}^2)$  and  $\mathbf{W}_k = \text{diag}(w_{1k}, \dots, w_{nk})$ . Updating block  $k$  with the others held fixed by one Fisher scoring step gives

$$\boldsymbol{\gamma}_k^{\text{new}} = (\mathbf{Z}_k^\top \mathbf{W}_k \mathbf{Z}_k + \lambda_k \mathbf{P}_k)^{-1} \mathbf{Z}_k^\top \mathbf{W}_k \mathbf{z}_k, \quad \mathbf{z}_k = \boldsymbol{\eta}_k + \mathbf{W}_k^{-1} \mathbf{u}_k, \quad (45.5.5)$$

the penalized version of the iteratively reweighted least squares of Theorem 34.4.1, with  $\mathbf{z}_k$  the working response for parameter  $k$ . Cycling over  $k = 1, \dots, K$  until nothing changes is backfitting (Theorem 44.2.1) applied to the  $K$  blocks.

- (c) **The normal location-scale model is orthogonal.** For [Eq. 45.5.2](#),

$$u_{i1} = \frac{y_i - \mu_i}{\sigma_i^2}, \quad w_{i1} = \frac{1}{\sigma_i^2}, \quad u_{i2} = \frac{(y_i - \mu_i)^2}{\sigma_i^2} - 1, \quad w_{i2} = 2, \quad (45.5.6)$$

and the expected cross-information  $-E(\partial^2 \ell_i / \partial \eta_{i1} \partial \eta_{i2})$  is zero. The two blocks are therefore orthogonal: neither Fisher scoring step needs the other's derivatives, the working response for  $\mu$  is  $\mathbf{z}_1 = \mathbf{y}$ , and the working response for  $\log \sigma$  is  $\mathbf{z}_2 = \boldsymbol{\eta}_2 + \frac{1}{2} \mathbf{u}_2$ .

*Proof.*

- (a) By the chain rule  $\partial \ell_i / \partial \boldsymbol{\gamma}_k = (\partial \ell_i / \partial \eta_{ik}) \partial \eta_{ik} / \partial \boldsymbol{\gamma}_k$ , and  $\partial \eta_{ik} / \partial \boldsymbol{\gamma}_k$  is the  $i$ th row of  $\mathbf{Z}_k$ . Summing  $u_{ik}$  times that row over  $i$  gives  $\mathbf{Z}_k^\top \mathbf{u}_k$ , and differentiating the quadratic penalty gives  $-\lambda_k \mathbf{P}_k \boldsymbol{\gamma}_k$ .
- (b) Differentiating [Eq. 45.5.4](#) once more, the contribution of observation  $i$  to the second derivative within block  $k$  is  $\partial^2 \ell_i / \partial \eta_{ik}^2$  times the outer product of the  $i$ th row of  $\mathbf{Z}_k$  with itself; taking expectations and summing over the independent responses gives  $-\mathbf{Z}_k^\top \mathbf{W}_k \mathbf{Z}_k - \lambda_k \mathbf{P}_k$ . A Fisher scoring step therefore reads

$$\boldsymbol{\gamma}_k^{\text{new}} = \boldsymbol{\gamma}_k + (\mathbf{Z}_k^\top \mathbf{W}_k \mathbf{Z}_k + \lambda_k \mathbf{P}_k)^{-1} (\mathbf{Z}_k^\top \mathbf{u}_k - \lambda_k \mathbf{P}_k \boldsymbol{\gamma}_k).$$

Adding and subtracting  $\mathbf{Z}_k^\top \mathbf{W}_k \mathbf{Z}_k \boldsymbol{\gamma}_k$  inside the bracket turns the right side into  $(\mathbf{Z}_k^\top \mathbf{W}_k \mathbf{Z}_k + \lambda_k \mathbf{P}_k)^{-1} \mathbf{Z}_k^\top \mathbf{W}_k (\mathbf{Z}_k \boldsymbol{\gamma}_k + \mathbf{W}_k^{-1} \mathbf{u}_k)$ , which is [Eq. 45.5.5](#).

- (c) With  $\ell_i = -\log \sigma_i - (y_i - \mu_i)^2 / (2\sigma_i^2) - \frac{1}{2} \log(2\pi)$ ,  $\mu_i = \eta_{i1}$  and  $\sigma_i = e^{\eta_{i2}}$ , direct

differentiation gives  $\partial \ell_i / \partial \eta_{i1} = (y_i - \mu_i) / \sigma_i^2$  and

$$\frac{\partial \ell_i}{\partial \eta_{i2}} = \sigma_i \frac{\partial \ell_i}{\partial \sigma_i} = \sigma_i \left\{ -\frac{1}{\sigma_i} + \frac{(y_i - \mu_i)^2}{\sigma_i^3} \right\} = \frac{(y_i - \mu_i)^2}{\sigma_i^2} - 1.$$

Differentiating again,  $\partial^2 \ell_i / \partial \eta_{i1}^2 = -1 / \sigma_i^2$ , which is already non-random, so  $w_{i1} = 1 / \sigma_i^2$ ; and  $\partial^2 \ell_i / \partial \eta_{i2}^2 = -2(y_i - \mu_i)^2 / \sigma_i^2$ , whose expectation is  $-2$ , so  $w_{i2} = 2$ . Finally  $\partial^2 \ell_i / \partial \eta_{i1} \partial \eta_{i2} = -2(y_i - \mu_i) / \sigma_i^2$ , with expectation zero because  $E(Y_i - \mu_i) = 0$ . The working responses follow from  $\mathbf{z}_k = \boldsymbol{\eta}_k + \mathbf{W}_k^{-1} \mathbf{u}_k$ : for  $k = 1$ ,  $z_{i1} = \mu_i + \sigma_i^2 (y_i - \mu_i) / \sigma_i^2 = y_i$ .

■

*Idea.* A GAMLSS fit is a stack of weighted smoothers. Each parameter's block is updated by the same penalized weighted least squares step that fitted a generalized additive model, with weights and working responses supplied by the log-likelihood of the chosen family. The rest is bookkeeping, which is why the framework accommodates any family whose derivatives can be written down.

Orthogonality in part (c) is special to this parameterization of the normal family; for most families the cross-information is not zero, and cycling over blocks then converges more slowly or must give way to a joint Newton step. The effective degrees of freedom of a block is the trace  $\text{tr}\{(\mathbf{Z}_k^\top \mathbf{W}_k \mathbf{Z}_k + \lambda_k \mathbf{P}_k)^{-1} \mathbf{Z}_k^\top \mathbf{W}_k \mathbf{Z}_k\}$  that Theorem 27.2.2 attached to ridge regression and Definition 30.1.1 to penalized least squares; the total of these traces goes into AIC (Proposition 29.2.4).

### 45.5.3 Choosing the family

The distribution is a modelling choice on a par with the link, made from what the response is. Positive with a right tail: gamma, inverse Gaussian, lognormal. Positive with adjustable skewness: the Box–Cox family, whose third parameter is a power estimated rather than assumed, generalizing Definition 22.2.1. Heavy tailed: a  $t$  family with the degrees of freedom as a third parameter. Counts with too many zeros: a zero-inflated or hurdle family (Definition 37.5.1). Proportions: a beta family reparameterized into a mean and a precision.

The ancestor is the LMS method of Cole and Green (1992) for growth charts: a Box–Cox power, a median and a coefficient of variation, each a smooth function of age — a three-parameter GAMLSS written a decade before the name, and still how paediatric reference charts are built.

### 45.5.4 Warnings

*Warning* (Identifiability). Each predictor in Eq. 45.5.1 contains an intercept and several smooth terms, and a constant moves freely between them. The remedy is that of Chapter 44: constrain every smooth term to sum to zero over the data (Definition 44.1.1), separately in every predictor. Without the constraints the penalized information is singular in as many directions as there are smooth terms. When the *same* covariate enters two predictors, the concurvity of Proposition 44.2.3 appears between parameters, and no *ceteris paribus* reading of a single parameter's curve survives.

*Warning* (The likelihood is unbounded). Let the mean predictor be flexible enough to interpolate one observation, say  $\mu(x_1) = y_1$ , and the scale predictor flexible enough to be small near  $x_1$  and unchanged elsewhere. The contribution of observation 1 to Eq. 45.5.3 is then  $-\log \sigma(x_1)$ , which grows without bound as  $\sigma(x_1) \rightarrow 0$  while the others stay finite: the unpenalized likelihood of a flexible location-scale model has no maximum. What stops the fit running away is the penalty, so the smoothing parameters are what make the problem well posed — and they cannot be chosen by maximizing the same likelihood.

### Example. Watching it happen

Fitting Eq. 45.5.2 to Engel's budgets with the scale penalty set to 0.01 instead of the chosen 10.0 raises the in-sample log-likelihood from 277.4 to 282.9, a gain a naive search would take. The ten-fold cross-validated log score — minus the mean predictive log density, so that smaller is better (Definition 45.6.1 and the remark in Section 45.6) — tells the truth: it moves from  $-1.1102$  to  $59.9$ . The wilder scale curve dips towards zero where the mean spline interpolates, and a held-out household near such a dip is assigned a density indistinguishable from zero.

## 45.5.5 A location-scale fit to Engel's budgets

### Example. Mean and scale together

Take Eq. 45.5.2 with a cubic B-spline basis of 12 columns for each predictor — knots at income quantiles, second-difference penalties (Definition 43.3.1) — and measure income and food expenditure in thousands of francs. The two smoothing parameters are chosen by ten-fold cross-validation of the log score over a grid, giving 31.6 for the mean and 10.0 for the scale, with effective degrees of freedom 8.75 and 5.59, 14.33 in all.

The fitted scale rises from 40.5 francs at an income of 500 to 184.5 at an income of 2000, a ratio of 4.55; the single constant the homoscedastic model fits is 107.4 francs, wrong in both directions at once. At an income of 1000 francs the fitted mean budget is 645.3 francs and the fitted ninetieth percentile 746.5. With the response in thousands of francs the AIC is  $-526.1$  against  $-368.0$  for the same mean spline with a constant scale, and the cross-validated log score  $-1.1102$  against  $-0.6938$ : most of what a homoscedastic model gets wrong here is the scale. The implied quantile curves  $\hat{\mu}(x) + \hat{\sigma}(x)z_\tau$  cannot cross, one clear advantage over the separate fits of Section 45.3.

```
import numpy as np
import statsmodels.api as sm
from scipy import stats
from scipy.interpolate import BSpline
data = sm.datasets.engel.load_pandas().data
DEGREE, N_INNER = 3, 8
CHOSEN = (10.0 ** 1.5, 10.0 ** 1.0)          # the pair the grid search picks
out
WILD = 0.01                                  # a scale penalty small enough to
run away

income = data["income"].to_numpy() / 1000.0    # thousands of francs
food = data["foodexp"].to_numpy() / 1000.0
```

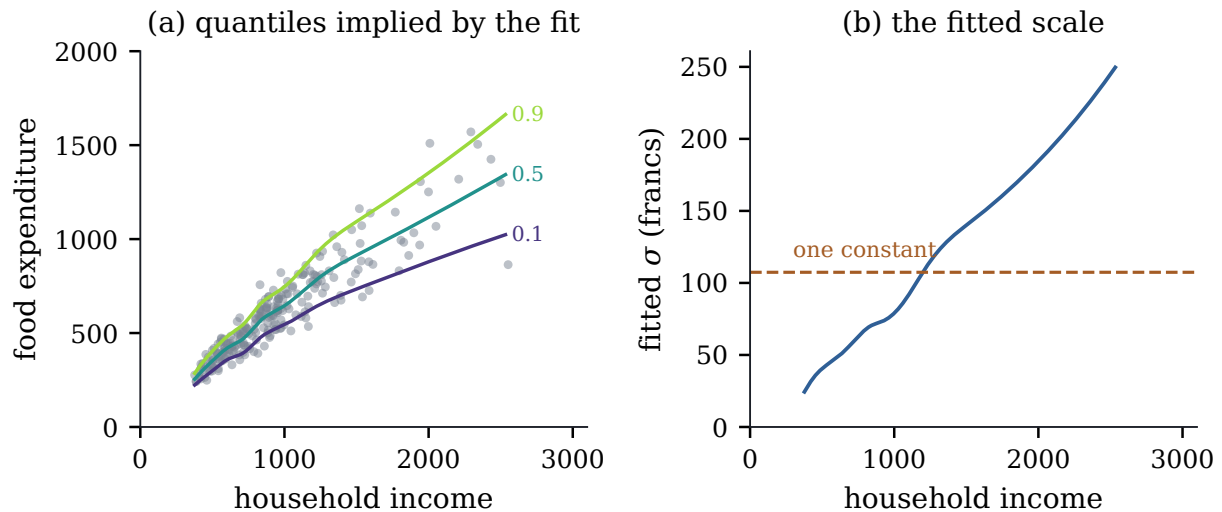


Figure 45.5.1: A normal location-scale GAMLSS for Engel's budgets, both predictors penalized splines in income. (a) The implied 0.1, 0.5 and 0.9 quantile curves. (b) The fitted scale against the homoscedastic constant.

```

n = len(food)

lo, hi = income.min(), income.max()
inner = np.quantile(income, np.linspace(0, 1, N_INNER + 2)[1:-1])
KNOTS = np.r_[[lo] * (DEGREE + 1), inner, [hi] * (DEGREE + 1)]

def basis(z):
    """The cubic B-spline basis of Section 43.3, with knots at income quantiles."""
    z = np.clip(np.atleast_1d(z), lo, hi)
    return np.asarray(BSpline.design_matrix(z, KNOTS, DEGREE).todense())

def difference_penalty(q, order=2):
    """ $K = D'D$  for the order-th difference matrix  $D$ : the penalty of Section 43.4."""
    D = np.diff(np.eye(q), order, axis=0)
    return D.T @ D

def gamlss_normal(Bm, Bs, y, lam_mu, lam_sig, iters=100, tol=1e-9):
    """Penalized Fisher scoring for  $y \sim N(\mu, \sigma^2)$ ,  $\mu = Bm \ b_\mu$ ,  $\log \sigma = Bs \ b_\sigma$ .

    One block at a time: the mean step is a penalized weighted least squares fit
    with
    weights  $1 / \sigma^2$ , and the scale step is a penalized Fisher scoring step whose
    expected information is the constant 2.
    """
    Km, Ks = difference_penalty(Bm.shape[1]), difference_penalty(Bs.shape[1])
    b_mu = np.linalg.solve(Bm.T @ Bm + lam_mu * Km, Bm.T @ y)
    b_sig = np.linalg.solve(Bs.T @ Bs, Bs.T @ np.full(len(y), np.log(y.std())))
    for _ in range(iters):
        sigma = np.exp(np.clip(Bs @ b_sig, -8, 4))
        w = 1.0 / sigma ** 2 # working weights for
        the mean
        new_mu = np.linalg.solve(Bm.T @ (w[:, None] * Bm) + lam_mu * Km, Bm.T @ (w *
        y))
        u = (y - Bm @ new_mu) ** 2 / sigma ** 2 - 1.0 # score in the log-
        sigma direction

```

```

        new_sig = b_sig + np.linalg.solve(2 * Bs.T @ Bs + lam_sig * Ks,
                                           Bs.T @ u - lam_sig * Ks @ b_sig)
        step = max(np.max(np.abs(new_mu - b_mu)), np.max(np.abs(new_sig - b_sig)))
        b_mu, b_sig = new_mu, new_sig
        if step < tol:
            break
    return b_mu, b_sig

def neg_log_score(mu, sigma, y):
    """Minus the mean log density: the logarithmic score of Section 45.6."""
    return -np.mean(stats.norm.logpdf(y, mu, sigma))

folds = np.random.default_rng(7).permutation(n) % 10

def cross_validate(design, lam_mu, lam_sig):
    """Ten-fold cross-validated log score of one specification."""
    total = 0.0
    for k in range(10):
        train = folds != k
        Bm, Bs = design(income[train])
        b_mu, b_sig = gamlss_normal(Bm, Bs, food[train], lam_mu, lam_sig)
        Bm, Bs = design(income[~train])
        total += neg_log_score(Bm @ b_mu, np.exp(np.clip(Bs @ b_sig, -8, 4)), food[~train])
    return total / 10

spline_scale = lambda z: (basis(z), basis(z))
Bm, Bs = spline_scale(income)
b_mu, b_sig = gamlss_normal(Bm, Bs, food, *CHOSEN)
mu, sigma = Bm @ b_mu, np.exp(Bs @ b_sig)
cv_chosen = cross_validate(spline_scale, *CHOSEN)
print(f"smoothing parameters: {CHOSEN[0]:.3g} for the mean, {CHOSEN[1]:.3g} for the
      scale")
print(f"cross-validated log score {cv_chosen:.4f}")

```

[Run this code online](#)

```

# barely penalize the scale: the in-sample likelihood climbs, the honest score
# collapses
b_mu_w, b_sig_w = gamlss_normal(Bm, Bs, food, CHOSEN[0], WILD)
wild_loglik = np.sum(stats.norm.logpdf(food, Bm @ b_mu_w, np.exp(np.clip(Bs @
    b_sig_w, -8, 4)))))
loglik = np.sum(stats.norm.logpdf(food, mu, sigma))
wild_cv = cross_validate(spline_scale, CHOSEN[0], WILD)
print(f"log-likelihood: {loglik:.1f} at the chosen penalty, {wild_loglik:.1f} at"
      f" lambda_sigma = {WILD:g}")
print(f"cross-validated log score: {cv_chosen:.4f} against {wild_cv:.4f}")

```

[Run this code online](#)

## 45.5.6 Variations

**Bayesian GAMLSS.** Every quadratic penalty is a normal prior on the coefficients (Proposition 39.5.1), so Eq. 45.5.3 is a log posterior up to the smoothing-parameter priors, and Section 39.4 transfers whole, with those parameters sampled rather than tuned (Klein, Kneib, Lang and Sohn 2015). **Boosting.** Feeding a GAMLSS log-likelihood to the componentwise boosting

of [Section 30.4](#) gives an algorithm that also selects which terms enter which predictor (Mayr, Fenske, Hofner, Kneib and Schmid 2012). **Multivariate responses.** With a multivariate family the dependence parameters become regression problems too; [Section 45.7](#) points at the literature.

## 45.5.7 Exercises

### 45.5.7.1 A. Check your understanding

#### Exercise. A1

In [Example \(Mean and scale together\)](#) the mean block has 12 basis columns and effective degrees of freedom 8.75. What are the two extreme values the effective degrees of freedom of that block could take, and which smoothing parameters produce them?

*Solution.* As  $\lambda_\mu \rightarrow 0$  the trace tends to 12, the number of columns; as  $\lambda_\mu \rightarrow \infty$  it tends to the dimension of the null space of the second-difference penalty, namely 2 (constants and straight lines). The fitted value 8.75 is much nearer the flexible end.

### 45.5.7.2 B. Practice

#### Exercise. B1

For a gamma response with mean  $\mu$  and shape  $\nu$ , density  $d(y) = \{\nu/\mu\}^\nu y^{\nu-1} e^{-\nu y/\mu} / \Gamma(\nu)$ , with log links on both parameters, compute  $u_{i1}$  and  $w_{i1}$  of [Theorem 45.5.1\(b\)](#) for the mean block. Is the model orthogonal in the sense of part (c)?

*Solution.* With  $\eta_1 = \log \mu$ ,  $\partial \ell / \partial \mu = \nu(y - \mu)/\mu^2$ , so  $u_1 = \mu \partial \ell / \partial \mu = \nu(y - \mu)/\mu$ . Then  $\partial^2 \ell / \partial \eta_1^2 = -\nu y / \mu$ , with expectation  $-\nu$ , so  $w_1 = \nu$ . The cross derivative  $\partial^2 \ell / \partial \eta_1 \partial \eta_2$  involves  $\nu(y - \mu)/\mu$  times  $\nu$ , whose expectation is zero, so the gamma model with these links is also orthogonal — a fact the algorithm exploits, and one that does not extend to three-parameter families.

#### Exercise. B2

Verify from [Eq. 45.5.6](#) that the working response for the scale block of the normal location-scale model is  $z_{i2} = \log \sigma_i + \frac{1}{2} \{(y_i - \mu_i)^2 / \sigma_i^2 - 1\}$ , and show that the scale step is therefore an ordinary penalized least squares smooth of the squared standardized residuals, shifted and halved.

### 45.5.7.3 C. Going deeper

#### Exercise. C1

Show that the normal family parameterized by  $(\mu, \sigma)$  is orthogonal for *any* increasing link on  $\sigma$ , not just the logarithm, and find a parameterization of the normal family in which it is not. What does orthogonality buy in [Theorem 45.5.1\(b\)](#)?